

Title: AI-Driven Adaptive Agile Software Development and Intelligent Project Management System

Abstract

Customer feedback contains valuable information about users' opinions, satisfaction levels, The **AI-Driven Adaptive Agile Software Development and Intelligent Project Management System** is an innovative software platform designed to improve the planning, monitoring, and management of Agile software projects using Artificial Intelligence. The system helps software teams manage project requirements, tasks, sprints, and team activities while adapting to changes throughout the software development lifecycle.

The proposed system uses AI techniques to analyze project data and provide intelligent recommendations such as task prioritization, effort estimation, risk identification, sprint planning, and progress prediction. It also supports Agile practices including product backlog management, sprint management, task assignment, progress tracking, and retrospective analysis.

The system follows the principles of **Adaptive Software Engineering**, allowing project plans and priorities to be dynamically adjusted based on changing requirements, project progress, and team feedback. By integrating AI with Agile software development practices, the system aims to reduce manual project management effort, improve decision-making, identify potential issues at an early stage, and enhance team productivity and collaboration.

The proposed solution demonstrates how modern AI technologies can be integrated with adaptive and Agile software engineering practices to create a flexible, intelligent, and efficient project management environment for software development teams.

Modern Tools and Technologies:

- Python
- FastAPI
- React.js
- Tailwind CSS
- PostgreSQL / MySQL
- Machine Learning / NLP
- Generative AI / LLM
- Git & GitHub
- REST APIs
- Agile / Scrum

Project Group Members

S.No.	Roll. No.	Student Name
1	2420030390	T. Siri
2	2420030514	Y. Anuhya
3	2420030530	D. Aarya Deeksha